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RESEARCH ARTICLE

MAG-KD: Mask-Guided Adaptive Gating Knowledge Distillation for Skeleton-Based Sign Language Recognition on Edge Devices

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ABSTRACT While artificial intelligence has significantly advanced sign language recognition (SLR), high-performing models often entail substantial computational complexity. This creates a significant barrier to deploying real-time applications on resource-constrained edge hardware. While knowledge distillation (KD) offers a potential solution, existing multi-teacher strategies are predominantly tailored for static images and remain unoptimized for skeleton-based graph convolutional networks (GCNs), which are inherently sensitive to pose estimation noise and require modeling complex temporal-semantic dependencies. To address this, we propose Mask-Guided Adaptive Gating Knowledge Distillation (MAG-KD), a framework specifically tailored for skeleton-based GCNs. Our approach adopts a decoupled distillation strategy, integrating a Mask-Guided Filtering Mechanism based on correctness-aware principles to strictly filter noisy supervision, and an Adaptive Gating Network leveraging context-aware features to dynamically synthesize teacher knowledge based on input complexity. Extensive evaluations across multiple random seeds demonstrate the superior stability of our approach. On the AUTSL dataset, our lightweight MAG-KD student (Student-M, 0.48M parameters) achieves a highly robust Top-1 mean accuracy of $94.25\% \pm 0.19$. Furthermore, our method proves strong cross-dataset generalization by achieving $66.33\% \pm 0.54$ on the WLASL-100 subset, significantly outperforming state-of-the-art KD methods. Comprehensive hardware profiling on an NVIDIA Jetson Nano confirms optimal edge feasibility: the deployed model achieves a real-time inference speed of 33.13 frames per second (FPS) while maintaining a highly efficient power profile of merely 6.23 W. Extended analysis indicates that the proposed algorithm operates optimally within the standard edge model regime, whereas static ensemble strategies remain more effective for ultra-compressed architectures with fewer than 100K parameters.

INDEX TERMS Computer vision, deep learning, edge computing, graph convolutional networks, knowledge distillation, sign language recognition.

I. INTRODUCTION

Sign language is the primary means of communication for the hearing-impaired community. With recent advances in AI, automated SLR has emerged as an active research direction to address this communication gap. Early SLR systems primarily relied on continuous RGB video streams

processed by computationally heavy 3D Convolutional Neural Networks (3D-CNNs) [1], [2], which incur prohibitive computational costs. To overcome this, a fundamental paradigm shift toward skeleton-based Graph Convolutional Networks (GCNs) has emerged as the standard, motivated by their extreme computational efficiency and robust invariance to complex background noise. The introduction of GCNs, such as ST-GCN [3] and 2s-AGCN [4], has revolutionized the field by effectively modeling the natural topological

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structure of human body movements. Building on this foundation, subsequent studies have progressively enhanced GCN architectures to capture more fine-grained spatiotemporal dependencies. Notable advancements include channel-wise topology refinement (CTR-GCN) [5], information-theoretic representation learning (InfoGCN) [6], and recent long-range architectures like SkateFormer [7]. These innovations have significantly pushed the boundaries of recognition accuracy, as highlighted in recent systematic reviews [8] and further corroborated by diverse structural exploration studies in skeleton-based recognition [9], [10], [11], [12], [13].

However, the continuous evolution of these models comes with a trade-off. State-of-the-art models often incorporate sophisticated co-occurrence mining mechanisms [14] or complex topology search, resulting in substantial parameter counts and floating point operations (FLOPs). This creates a significant bottleneck when deploying SLR systems on resource-constrained edge devices such as smart glasses and service robots, where real-time processing and low power consumption are mandatory. To address this hardware barrier [15], Knowledge Distillation (KD) [16] has established itself as a critical compression paradigm. The methodology has evolved from single-teacher feature distillation (e.g., FitNets [17], Contrastive Distillation [18]) to advanced multi-teacher frameworks. Recent approaches utilize dynamic weighting mechanisms—such as Adaptive Ensemble KD (AEKD) [19], Decoupled KD (DKD) [20] and reinforcement learning-based selection [21]—to prioritize reliable teachers, achieving notable success in static visual recognition [22]. Alongside these, various structural and confidence-based regularization strategies have also been extensively explored to optimize the student's learning capacity [23], [24], [25], [26], [27].

Despite these successes, most existing methods are tailored for image data, leaving skeleton-based GCNs underexplored. While there are initial attempts at Graph KD, such as LSP [28] for local structure preservation, FreeKD [29] for free-direction distillation, and recent Spiking GNNs [30], they often face challenges specific to SLR. Sign language relies heavily on the semantic motion of hands and face over time, which differs significantly from static graphs. Furthermore, these graph-specific distillation techniques still inherit a critical vulnerability when applied to real-world edge scenarios: the susceptibility to dynamic pose tracking errors.

A. IMPACT OF POSE ESTIMATION ERRORS

Despite the theoretical success of KD, directly transplanting these methods to skeleton-based SLR presents a critical challenge. Skeleton-based GCNs fundamentally depend on the quality of input skeletons extracted by upstream pose estimation frameworks (e.g., HRNet [31], MediaPipe [32]). In real-world sign language scenarios characterized by rapid hand movements and self-occlusions, these upstream extractors inevitably introduce severe topological

noise—including missing joints, spatial jitter, and inverted coordinates. Consequently, when heavy GCN architectures process these corrupted graphs, they actively amplify the upstream errors [33], [34], leading to wildly inaccurate predictions.

B. THE ORACLE ASSUMPTION AND NEGATIVE TRANSFER

Crucially, existing multi-teacher distillation strategies predominantly operate on the “oracle assumption”—presuming that teachers always provide correct supervision. However, recent literature critically challenges this assumption. Specifically, Lan et al. [35] demonstrated that well-trained teachers still contain incorrect predictions that contradict the ground truth, causing severe student confusion. Furthermore, Zhao [36] empirically investigated how noisy teacher labels lead to sub-optimal distillation. Most notably, a highly recent study by Mason-Williams et al. [37] reveals that KD can disproportionately favor the transfer of a teacher's incorrect predictions. This systematic error amplification is formally recognized as “negative transfer” [38], a detrimental phenomenon where blindly forcing a student to mimic a corrupted teacher actively degrades its performance in real-world deployment.

C. THEORETICAL DIFFERENTIATION: DECOUPLED GATING AND HARD MASKING

To overcome the limitations of negative transfer in noisy skeleton graphs, standard adaptive methods are insufficient. First, unlike standard Decoupled KD (DKD) [20], which only decouples target and non-target classes within the logit space of a *single* teacher, our framework uniquely decouples the *feature* and *logit* distillation lanes across *multiple* heterogeneous teachers. Features require stable aggregation to learn robust structural priors, while logits demand strict filtering. Second, while traditional confidence-aware methods [39] utilize soft probabilities (which can still leak noisy gradient guidance if an incorrect teacher is overconfident), our method employs a deterministic Hard Correctness Mask. This deterministic gating ensures the absolute truncation of negative transfer in inherently noisy skeleton-based settings.

To overcome these limitations, we propose MAG-KD, a framework specifically designed for skeleton-based SLR on edge devices. Our proposed pipeline seamlessly bridges the gap between high-performance training and efficient edge deployment. Crucially, we adopt a decoupled distillation strategy to balance stability and precision. While intermediate features are distilled via static averaging to preserve robust structural representations, the final logits are strictly regulated by our MAG module. Specifically, this module employs a Mask-Guided Filtering Mechanism operating on a correctness-aware principle to act as a hard filter against noisy supervision, and an Adaptive Gating Network leveraging context-aware features to dynamically weigh the remaining reliable contributions. Subsequently, the

optimized student model is transitioned to the deployment phase, where it is converted to ONNX format and accelerated via a TensorRT engine with FP16 quantization. This end-to-end workflow ensures that the final model achieves real-time performance on the NVIDIA Jetson Nano while retaining the high accuracy of heavy teachers.

The contributions are threefold:

(1) We propose MAG-KD, a noise-robust distillation framework that uniquely decouples feature and logit distillation across multiple heterogeneous teachers to effectively mitigate negative transfer caused by pose estimation errors.

(2) We introduce a novel Mask-Guided Filtering Mechanism utilizing a deterministic hard mask, integrated with a Context-aware Adaptive Gating Network to dynamically synthesize reliable teacher knowledge.

(3) Extensive experiments demonstrate that our method achieves state-of-the-art performance on the AUTSL and WLASL datasets while maintaining real-time efficiency on edge devices.

II. METHODOLOGY

A. PRELIMINARIES

1) GRAPH-BASED SKELETON REPRESENTATION

Following standard conventions in skeleton-based action recognition [3], we represent a sign language video as a spatio-temporal graph $G = (\mathcal{V}, \mathcal{E})$. The node set $\mathcal{V} = \{v_{ti} \mid t = 1, \dots, T; i = 1, \dots, N\}$ comprises $N = 27$ body keypoints across T frames, covering the upper body and hand joints crucial for sign language. The edge set $\mathcal{E}$ consists of two subsets: intra-frame edges connecting joints based on natural human connectivity (spatial dimension) and inter-frame edges connecting the same joints across consecutive frames (temporal dimension). The input feature tensor is denoted as $\mathbf{X} \in \mathbb{R}^{N \times C \times T}$, where C represents the channel dimension, representing 2D/3D coordinates and confidence scores.

2) SPATIAL-TEMPORAL GRAPH CONVOLUTION

Since our framework is backbone-agnostic (allowing various architectures like ST-GCN [3], CTR-GCN [5]), we briefly formulate the general graph convolution operation used to extract features. For the spatial dimension at a given time step, the feature update rule is defined as [3]:

$$\mathbf{H}^{(l+1)} = \sigma \left(\sum_j \tilde{\mathbf{D}}_j^{-\frac{1}{2}} \tilde{\mathbf{A}}_j \tilde{\mathbf{D}}_j^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}_j^{(l)} \right), \quad (1)$$

where $\mathbf{H}^{(l)}$ is the feature map at layer l (with $\mathbf{H}^{(0)} = \mathbf{X}$); $\mathbf{A}_j$ denotes the adjacency matrix representing the physical connections in the j -th spatial partition strategy; $\tilde{\mathbf{A}}_j = \mathbf{A}_j + \mathbf{I}$ includes self-loops; $\tilde{\mathbf{D}}_j$ is the diagonal degree matrix for normalization; $\mathbf{W}_j$ is the learnable weight matrix, and $\sigma(\cdot)$ is the activation function. In the temporal dimension, standard temporal convolutions (TCN) are applied to model motion dynamics. The final output is a probability distribution $\hat{\mathbf{y}} \in \mathbb{R}^K$ over K classes.

3) MULTI-TEACHER KNOWLEDGE DISTILLATION FORMULATION

In a general multi-teacher KD setting, a student network S learns from a pool of M teacher networks $\{T_1, \dots, T_M\}$. The objective function is typically defined as a weighted sum of supervision signals:

$$\mathcal{L}_{\text{MTKD}} = \mathcal{L}_{\text{CE}} + \sum_{m=1}^M w_m \cdot \mathcal{L}_{\text{KD}}(T_m, S), \quad (2)$$

where w_m represents the importance weight of the m -th teacher. Existing methods like AVER [40] simply set static weights $w_m = 1/M$. More advanced approaches like MTKD-RL [21] attempt to learn dynamic weights via reinforcement learning. However, these methods often couple the weights for both logit and feature distillation (i.e., assuming $w_m^{\text{logit}} \equiv w_m^{\text{feat}}$).

B. THE PROPOSED MAG-KD STRATEGY

To address the challenges of noisy skeleton inputs and resource-constrained deployment, we propose MAG-KD, an end-to-end multi-teacher distillation framework. As illustrated in Fig. 1, the system architecture is structured into two distinct operational stages: the *Training Phase* (top section) and the *Inference Phase* (bottom section).

Execution Workflow: To make the methodology highly accessible, the end-to-end pipeline operates in four sequential steps: (1) *Inputs:* The 2D skeleton sequences are fed simultaneously into both the frozen heterogeneous teachers and the trainable student. (2) *Logit-Context Gating:* The intermediate 1D features and final logits are extracted to form a unified context vector, which the Adaptive Gating Network uses to generate raw importance scores. (3) *Correctness Masking:* The Mask-Guided Filter applies a deterministic hard mask (derived from ground-truth targets) to completely nullify erroneous teacher guidance, refining the scores into dynamic weights. (4) *Decoupled Distillation:* Finally, structural knowledge is transferred via static feature averaging, while precise decision boundaries are transferred via the dynamically weighted logit distillation.

1) PHASE 1: DUAL-BRANCH FEATURE EXTRACTION

In the top section of Fig. 1, the training workflow is organized into two parallel branches. On the left, the *Teacher Branch* comprises a frozen, heterogeneous ensemble (SL-GCN [41], CTR-GCN [5], TD-GCN [42]) to provide diverse supervision. On the right, the *Student Branch* features the lightweight STA-GCN [43], which is the target for optimization. Both branches process the input skeletons simultaneously to extract intermediate feature vectors and generate class logits, feeding them into the central distillation block.

2) PHASE 2: DECOUPLED DISTILLATION STRATEGY

Bridging the Teacher and Student branches is the central MAG-KD framework. Unlike traditional methods that

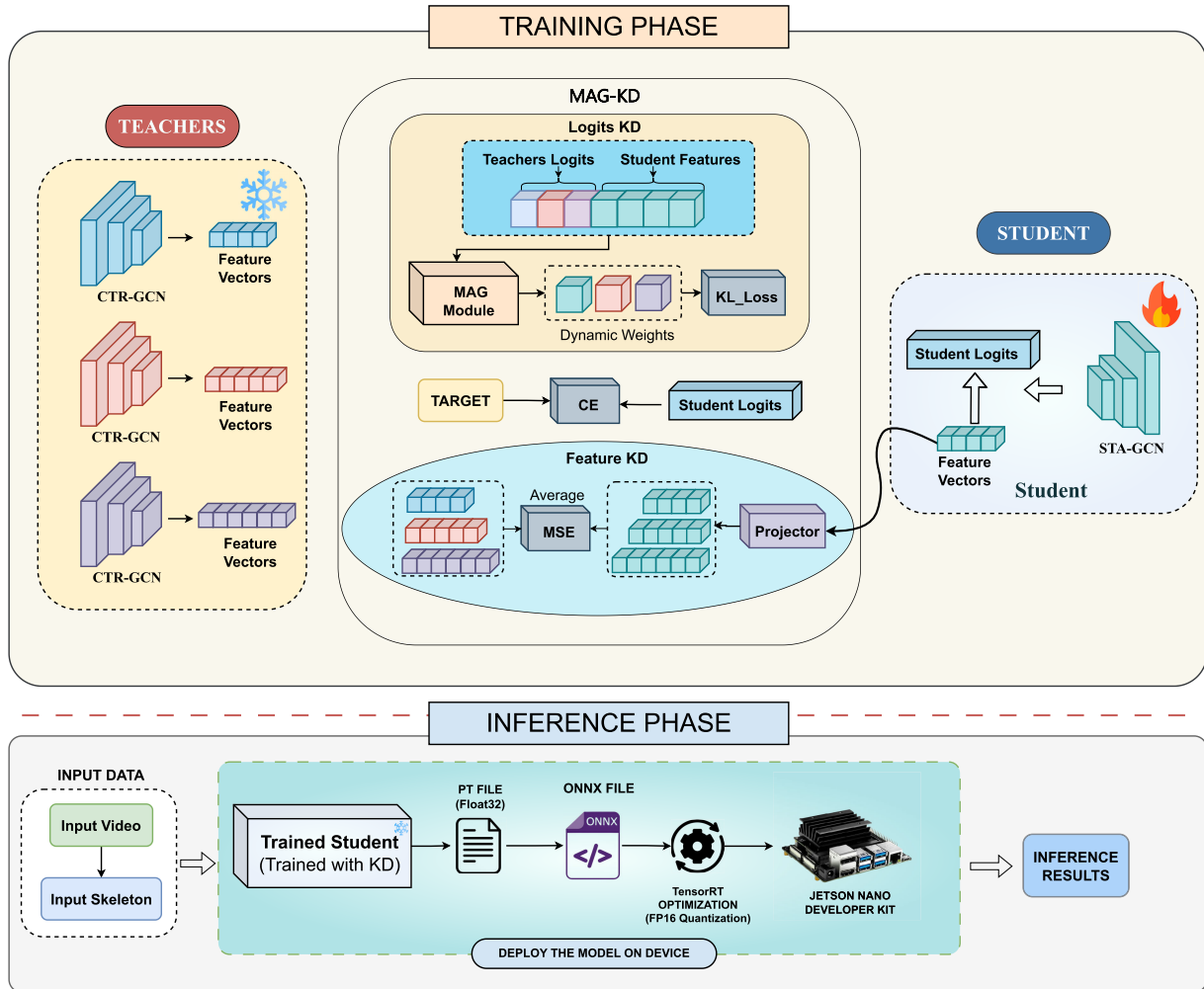


FIGURE 1. Overview of the MAG-KD Framework. The architecture is explicitly divided into two phases. The top block illustrates the **Offline Training phase**, where the pipeline adopts a decoupled distillation strategy regulated by the MAG and Projector modules. The heterogeneous Teacher ensemble (SL-GCN, CTR-GCN, TD-GCN) is frozen to provide diverse supervision. The distillation process is split into a Feature Alignment Path (using static averaging and MSE) and a Logit Distillation Path (using the MAG Module to dynamically filter noisy predictions). The bottom block depicts the **Online Inference phase**, where the optimized student is completely decoupled from the heavy teachers, exported to ONNX, and accelerated via TensorRT for edge deployment.

couple distillation objectives, we implement a *decoupled strategy* that splits the knowledge transfer into two specific pathways:

- 1) *Feature Alignment Path*: Located at the bottom of the central block, this path prioritizes structural stability for intermediate representations. Auxiliary Projectors align the Student's lower-dimensional features with the Teachers' space, followed by a static averaging strategy to minimize the Mean Squared Error (MSE).
- 2) *Logit Distillation Path*: Located at the top of the central block, this path prioritizes precision for decision boundaries. The Teachers' logits and Student's features are fed into the MAG Module, which dynamically generates refined weights to regulate the KL Divergence loss, filtering out negative transfer.

3) PHASE 3: THE MAG MODULE MECHANISM

Inside the MAG Module (detailed in Fig. 2), the system performs dynamic weight generation. It first aggregates the Teachers' logits and Student's features into a unified context. This context is processed by the Adaptive Gating Network (AGN) to produce raw importance scores. To ensure reliability, these scores are strictly filtered by a *Correctness-aware Mask*, effectively zeroing out contributions from Teachers that fail to predict the correct class, before being normalized for the final loss calculation.

4) PHASE 4: OPTIMIZATION AND EDGE DEPLOYMENT

The bottom section of Fig. 1 depicts the deployment workflow, which is decoupled from the training loop. Once the Student model is fully optimized via the combined CE,

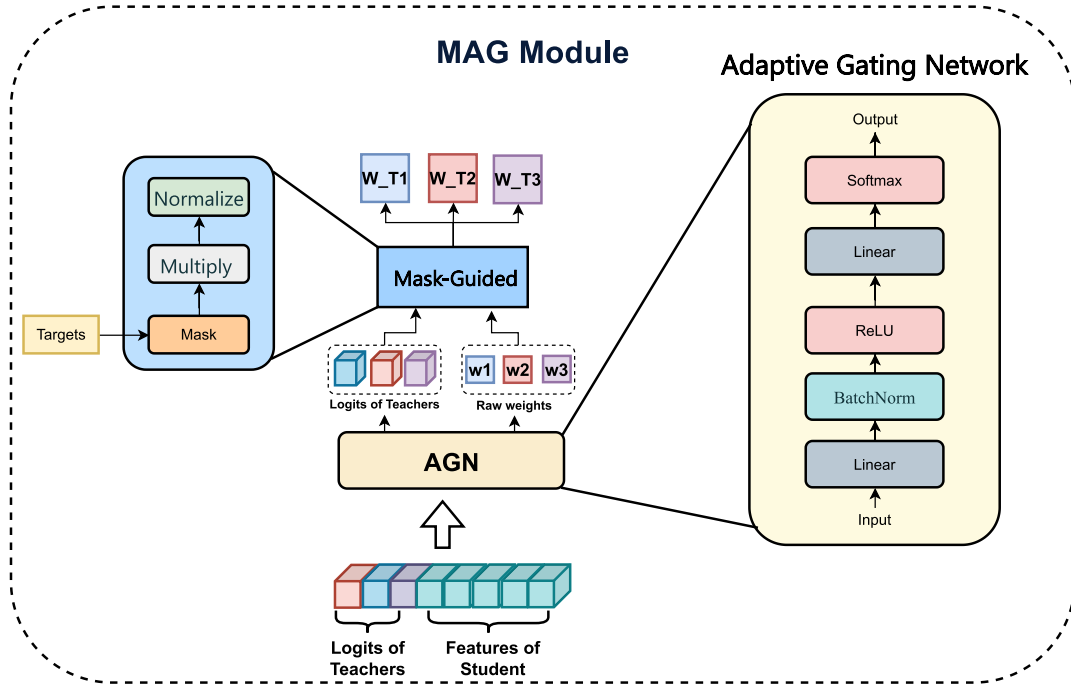


FIGURE 2. Internal Structure of the MAG Module. (Right) The Adaptive Gating Network (AGN) utilizes an MLP-based architecture (Linear-BN-ReLU-Linear-Softmax) to compute raw weights from the concatenated context (Teacher Logits + Student Features). (Left) The Mask-Guided Filter refines these weights by comparing Teachers' predictions (Logits) with Ground Truth Targets via a Mask-Multiply-Normalize mechanism, ensuring the Student only learns from correct supervision.

MSE, and KL losses, the heavy Teacher ensemble and MAG modules are discarded. The trained Student is then exported to ONNX format and optimized using TensorRT (FP16 quantization). Finally, the accelerated engine is deployed on the NVIDIA Jetson Nano to achieve real-time inference.

C. MASK-GUIDED FILTERING MECHANISM

In skeleton-based SLR, input data extracted from pose estimators often suffers from jittering or missing joints. Consequently, even high-performing Teacher models may produce incorrect predictions on noisy samples. Blindly distilling knowledge from these erroneous predictions leads to *negative transfer*, degrading the Student's performance.

Negative Transfer Rationale: We explicitly define this failure mode in skeleton settings: when upstream pose estimators produce corrupted graphs (e.g., due to self-occlusions), teacher networks processing these graphs will generate wildly inaccurate logits. Standard distillation blindly forces the student to mimic these confident but incorrect distributions—a detrimental phenomenon recognized as negative transfer. Our hard correctness mask acts as a deterministic gatekeeper, completely severing the gradient flow from teachers when their Top-1 prediction contradicts the ground truth, thereby immunizing the student against harmful supervision.

To mitigate this, we propose a hard-filtering mechanism called Correctness-aware Masking (depicted as the Filter in Fig. 2). This module evaluates the reliability of each

teacher T_m for a given input sample x_i by comparing the teacher's prediction against the ground truth label y_i . The binary mask $M_{m,i}$ is defined as:

$$M_{m,i} = \mathbb{I} \left(\arg \max(\mathbf{z}_{T_m}^{(i)}) = y_i \right), \quad (3)$$

where $\mathbf{z}_{T_m}^{(i)} \in \mathbb{R}^K$ denotes the logit vector of the m -th teacher, and $\mathbb{I}(\cdot)$ is the indicator function which equals 1 if the condition holds and 0 otherwise. This mask acts as a strict gate: if a teacher fails to classify the sample correctly, its contribution to the logit distillation loss is completely nullified, resulting in $M_{m,i} = 0$.

D. ADAPTIVE GATING NETWORK

While the Masking module (Section II-C) filters out erroneous teachers, the remaining “good” teachers may still vary in their expertise regarding specific sign gestures. To dynamically assign importance weights, we incorporate the Adaptive Gating Network (AGN) within the MAG module.

1) CONTEXT AGGREGATION AND WEIGHT GENERATION

The gating process initiates by constructing a context vector $\mathbf{c}_i$ that encapsulates both the student's current representation and the consensus among teachers. Let $\mathbf{f}_S^{(i)}$ denote the global average pooled feature of the student and $\mathbf{z}_T^{(i)}$ be the concatenated logits of all teachers. The context is

formulated as:

$$\mathbf{c}_i = [\mathbf{f}_S^{(i)} \oplus \mathbf{z}_{T_1}^{(i)} \oplus \cdots \oplus \mathbf{z}_{T_M}^{(i)}], \quad (4)$$

where $\oplus$ denotes concatenation. Subsequently, as illustrated in the right block of Fig. 2, the AGN processes this context vector using a lightweight MLP to generate raw importance scores. The architecture comprises a sequence of Linear, Batch Normalization (BN), ReLU activation, and a final Softmax layer:

$$\mathbf{w}_i^{\text{raw}} = \text{Softmax}(\mathbf{W}_2 \cdot \sigma(\text{BN}(\mathbf{W}_1 \cdot \mathbf{c}_i))), \quad (5)$$

where $\mathbf{W}_1, \mathbf{W}_2$ are learnable parameters and σ is the ReLU function.

2) MASK-MULTIPLY-NORMALIZE MECHANISM

To strictly enforce the correctness constraint, the generated raw weights are refined by the binary mask M derived in (3). The final adaptive weight $\hat{w}_{m,i}$ for the m -th teacher is computed via a renormalization step:

$$\hat{w}_{m,i} = \frac{w_{m,i}^{\text{raw}} \cdot M_{m,i}}{\sum_{j=1}^M w_{j,i}^{\text{raw}} \cdot M_{j,i} + \epsilon} \quad (6)$$

where ϵ is a small constant to prevent division by zero. This mechanism ensures that the total weight sums to 1 while effectively nullifying the contribution of unreliable teachers ($M_{m,i} = 0$).

E. OPTIMIZATION OBJECTIVE

The total training objective $\mathcal{L}_{\text{total}}$ combines the standard classification loss with our decoupled distillation losses. Unlike prior works that use the same weights for all distillation terms, we employ a decoupled weighting strategy:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{KD}}^{\text{logit}} + \beta \mathcal{L}_{\text{KD}}^{\text{feat}} \quad (7)$$

1) RATIONALE AGAINST LAYER-WISE DISTILLATION

We strategically perform knowledge distillation exclusively at the final 1D globally pooled feature layer (penultimate layer) and the logits layer, deliberately bypassing intermediate layer-wise alignment. Because our multi-teacher pool consists of heterogeneous GCN architectures (e.g., SL-GCN, TD-GCN) employing fundamentally different structural graph topologies and adjacency matrices, forcing the lightweight student to mimic intermediate feature maps would cause severe topological confusion. Distilling only the condensed 1D representations allows the student to absorb the optimal semantic knowledge while strictly avoiding structural overfitting.

1. Task Loss ($\mathcal{L}_{\text{CE}}$): The standard Cross-Entropy loss between the Student's prediction $\mathbf{z}_S$ and ground truth $\mathbf{y}$.

2. Logit Distillation Loss ($\mathcal{L}_{\text{KD}}^{\text{logit}}$): This term utilizes the dynamic weights $\hat{w}_{m,i}$ from the MAG module (6) to transfer precise decision boundaries:

$$\mathcal{L}_{\text{KD}}^{\text{logit}} = \sum_{m=1}^M \hat{w}_{m,i} \cdot \tau^2 \mathcal{D}_{\text{KL}} \left(\sigma \left(\frac{\mathbf{z}_{T_m}}{\tau} \right), \sigma \left(\frac{\mathbf{z}_S}{\tau} \right) \right), \quad (8)$$

where $\mathcal{D}_{\text{KL}}$ is the KL Divergence, σ is the softmax function, and τ is the temperature parameter.

3. Feature Alignment Loss ($\mathcal{L}_{\text{KD}}^{\text{feat}}$): Since intermediate features contain general structural knowledge, we utilize a static averaging strategy where weights are set to $1/M$ rather than strict filtering. We employ auxiliary projectors ϕ_m to align dimensions and minimize the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{KD}}^{\text{feat}} = \frac{1}{M} \sum_{m=1}^M \|\phi_m(\mathbf{F}_S) - \mathbf{F}_{T_m}\|^2. \quad (9)$$

III. EXPERIMENTS

A. DATASET AND EVALUATION PROTOCOL

We evaluate our proposed framework on the AUTSL dataset [44], a large-scale benchmark for Turkish Sign Language recognition collected using Microsoft Kinect V2. The dataset contains 38,336 video samples performed by 43 signers across 20 different background environments, covering 226 sign classes. Following the standard protocol, we adopt the predefined train/test split provided by the dataset authors.

To evaluate the generalizability and language-agnostic properties of our approach, we also conduct experiments on the Word-Level American Sign Language (WLASL) dataset [45]. Specifically, we utilize the WLASL-100 subset, which contains 100 classes with 1442 training and 596 validation video samples. This dataset presents severe real-world challenges such as extreme class imbalance, varying illumination, and high intra-class variations.

B. EXPERIMENTAL SETTINGS

We configure the model by employing a heterogeneous ensemble of three GCN models as the Teacher network: CTR-GCN [5], SL-GCN [41], and TD-GCN [42]. The Teachers are pre-trained on the target dataset and frozen during the distillation process, while the Student network is the lightweight STA-GCN [43]. To align the feature dimensions between the Student and Teachers, the auxiliary Projectors are implemented as simple two-layer MLPs with ReLU activation. In terms of implementation details, all training experiments are conducted using the PyTorch 2.1.2 framework with CUDA 12.1 on a workstation equipped with a single NVIDIA RTX A4000 GPU. The Student network is trained for 150 epochs using the Stochastic Gradient Descent (SGD) optimizer with Nesterov momentum of 0.9, a weight decay of 10^{-4} , and a batch size of 64; the initial learning rate is set to 0.1 and decays by a factor of 10 at epochs 60, 90, and 120 for both the baseline and the proposed MAG-KD to ensure a strictly fair comparison.

To provide unambiguous definitions of the network capacities utilized in our framework, Table 1 summarizes the physical parameters, floating-point operations (FLOPs), and baseline accuracies of all teacher models and student variants. Specifically, Student-M (Medium) serves as our primary target for the proposed MAG-KD framework,

TABLE 1. Model cast: Capacity and baseline performance of teachers and student variants evaluated on the AUTSL test set.

Model Role & Architecture	Params (M)	FLOPs (G)	Acc (%)
Teacher 1: SL-GCN	4.87	3.53	95.35
Teacher 2: TD-GCN	1.42	2.09	94.92
Teacher 3: CTR-GCN	1.49	2.20	94.90
Student-M (STA-GCN)	0.48	0.55	91.96 $\pm$ 0.29
Student-S (STA-GCN)	0.07	0.08	83.57

while Student-S (Small) represents an ultra-compressed variant designed for extreme edge scenarios.

Finally, regarding the hyperparameters for the knowledge distillation objectives, we empirically set the distillation temperature $\tau = 4$, while the loss balancing coefficients are configured as $\alpha = 1.0$ for the logit-based distillation and $\beta = 1.0$ for the feature alignment loss. To ensure statistical robustness and mitigate the variance of network initialization, the random seed is set to 1, 42, and 100 across different experimental runs.

Regarding the training overhead, integrating the MAG module introduces a marginal increase of 0.439M parameters and 0.0004G FLOPs. The training time per epoch increases from 82.5 seconds (baseline) to 188 seconds due to the dynamic gating computation. However, this overhead is entirely acceptable since it strictly occurs during the offline training phase and does not affect the inference speed of the deployed student model.

1) EDGE DEPLOYMENT SETUP

To evaluate the real-world feasibility, we deploy the optimized models on an NVIDIA Jetson Nano Developer Kit 4GB RAM [46], configured in 10 W (MAXN) power mode.

During the inference phase, the system takes an extracted 2D skeleton sequence tensor as input, strictly formatted as [Batch, Channels, Frames, Joints, Persons], and outputs the predicted gesture class probability distribution. The deployment workflow, corresponding to the deployment phase illustrated in Fig. 1, involves first exporting the trained PyTorch models to the intermediate ONNX format, and subsequently compiling them into TensorRT engines [47] using the `trtexec` command-line utility. We specifically utilize FP16 precision for all deployments. Preliminary experiments indicated that INT8 quantization on the Jetson Nano's Maxwell architecture does not yield significant throughput gains while introducing non-negligible accuracy degradation; thus, FP16 was selected as the optimal balance for this specific edge hardware.

We measure the performance using the following metrics:

- **Latency and Throughput:** Measured end-to-end with a batch size of 1, averaged over 200 iterations after a 20-iteration warm-up phase.
- **Power Consumption:** The average system power input (POM_5V_IN) is monitored using the `tegrastats` utility during the inference loop.

- **Energy Efficiency:** Defined as the number of frames processed per Watt (FPS/W), reflecting the model's suitability for battery-constrained environments.
- **Hardware Utilization:** We record the peak GPU utilization (%) and system RAM usage (MB) to assess the resource footprint.
- **Model Size:** Reported as the physical disk size of the deployed TensorRT engine file.

C. COMPARISON WITH STATE-OF-THE-ARTS

In this section, we compare our method with state-of-the-art multi-teacher knowledge distillation approaches on the large-scale AUTSL benchmark dataset [44]. The quantitative results are presented in Table 2. To comprehensively evaluate the proposed MAG-KD, we report experimental results comparing three distinct groups: (1) the teacher pool, consisting of high-capacity models (SL-GCN [41], TD-GCN [42], CTR-GCN [5]) that represent the upper bound of performance; (2) the student baseline, which is the lightweight STA-GCN [43] trained from scratch without distillation; and (3) various multi-teacher KD methods. These methods include traditional logit averaging (AVER [16]), feature-based alignment (FitNet [17]), and graph-specific distillation techniques such as Local Structure Preservation (LSP [28]). We also include recent adaptive mechanisms like AEKD [19], CA-MKD [39], and MTKD-RL [21]. We also report the number of parameters and FLOPs to highlight the trade-off between accuracy and computational efficiency. To guarantee statistical validity, the results of the baseline and our method are reported as Mean $\pm$ Standard Deviation across three random seeds.

D. EVALUATION ON WLASL DATASET

To verify that our proposed MAG-KD framework is language-agnostic and generalizes well beyond the Turkish Sign Language domain, we conduct additional experiments on the WLASL-100 dataset. As presented in Table 3, the baseline STA-GCN achieves a limited accuracy of 54.92% $\pm$ 0.77%. By applying MAG-KD, the student's performance is significantly boosted to 66.33% $\pm$ 0.54%, achieving an absolute improvement of +11.41% and outperforming all other state-of-the-art multi-teacher KD baselines. This substantial gain provides strong evidence that our masking-based distillation design effectively adapts to different linguistic topologies and dataset scales without requiring language-specific heuristics.

E. ABLATION STUDIES AND DEEP ANALYSIS

To validate the contribution of each component, we conduct an ablation study detailed in Table 4. First, regarding the impact of the Correctness Mask, the results demonstrate that eliminating this filter leads to a sharp performance decline of 8.95%. This confirms its critical role in preventing negative transfer by strictly blocking noisy supervision from erroneous teachers.

TABLE 2. Comparison of Top-1 accuracy of various knowledge distillation methods on the AUTSL dataset. To ensure statistical robustness, the baseline and our proposed MAG-KD results are reported as Mean $\pm$ Std across 3 random seeds (1, 42, 100).

Network	Params (M)	FLOPs (G)	Method	Acc@1 (%)
<i>Teacher Pool</i>				
SL-GCN	4.87	3.53	Teacher (Pretrained)	95.35
TD-GCN	1.42	2.09	Teacher (Pretrained)	94.92
CTR-GCN	1.49	2.20	Teacher (Pretrained)	94.90
<i>Student</i>				
STA-GCN	0.48	0.55	Baseline	91.96 $\pm$ 0.29
			MTKD-RL [21]	90.25
			AVER [16]	91.23
			Single Best (KD)	91.88
			AVER (KD+FitNet)	92.17
			AEKD [19]	93.19
			CA-MKD [39]	93.37
			LSP [28]	93.57
			MAG-KD (Ours)	94.25 $\pm$ 0.19

TABLE 3. Performance comparison of various knowledge distillation methods on the WLASL-100 dataset. The baseline and our proposed method report Mean $\pm$ Std over 3 random seeds (1, 42, 100).

Network	Params (M)	FLOPs (G)	Method	Acc@1 (%)
<i>Teacher Pool</i>				
SL-GCN	4.84	3.53	Teacher (Pretrained)	71.64
TD-GCN	1.39	2.09	Teacher (Pretrained)	65.10
CTR-GCN	1.46	2.20	Teacher (Pretrained)	63.93
<i>Student</i>				
STA-GCN	0.45	0.55	Baseline	54.92 $\pm$ 0.77
			CA-MKD [39]	62.08
			Single Best (KD)	64.43
			AVER [16]	64.60
			AEKD [19]	65.44
			AVER (KD + Feat)	65.94
			LSP [28]	66.11
			MTKD-RL [21]	66.28
			MAG-KD (Ours)	66.33 $\pm$ 0.54

TABLE 4. Ablation study analyzing the impact of the correctness mask on the AUTSL dataset.

Method Variation	Logits	Features	Mask Type	Acc@1 (%)	Δ
Gated Feature	Gated	Gated	Hard	93.00	-1.25
No Mask	Gated	Average	None	85.30	-8.95
Soft Masking	Gated	Average	Soft	94.20	-0.05
MAG-KD (Ours)	Gated	Average	Hard	94.25	-

Furthermore, we experiment with a Soft Masking variant where the gating weights are normalized via softmax instead of a hard binary threshold. As shown in Table 4, Soft Masking achieves 94.20%, which is marginally inferior to the Hard Masking's 94.25%. The hard threshold strictly severs noisy logit flows, proving to be slightly more effective at preventing negative transfer than soft down-weighting.

1) FEATURE VARIANCE ANALYSIS

We evaluate the Feature Strategy by comparing our proposed Hybrid approach against a fully adaptive method where features are also gated. Dynamic feature gating reduces performance by 1.25%. To defend the static averaging choice for features, we calculate the feature variance within the

ensemble. The variance of the highest-performing single teacher (SL-GCN) is 1.247, whereas the static fused feature variance drops to 0.342. This profound reduction indicates that static averaging inherently stabilizes the structural representation, rendering dynamic gating redundant and mathematically sub-optimal at the feature level.

2) VALIDATION LOSS CURVE

As depicted in the validation loss curves (see Fig. 3), the proposed MAG-KD framework converges significantly smoother and avoids the late-stage overfitting spikes observed in the baseline model, explicitly highlighting the regularization capability of our masked ensemble approach.

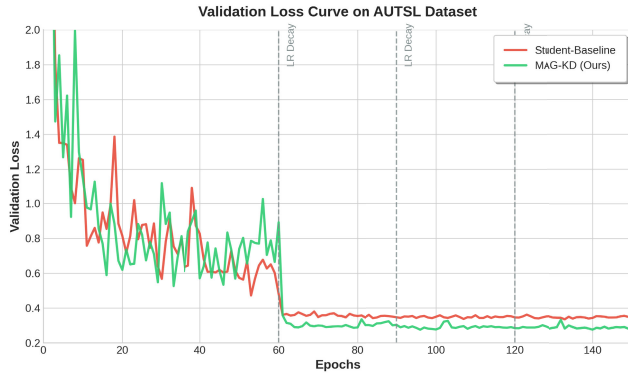


FIGURE 3. Validation loss curve comparison between the standalone Baseline and MAG-KD on the AUTSL dataset. MAG-KD exhibits smoother convergence and better regularization.

3) ROBUSTNESS TO SYNTHETIC POSE NOISE

To empirically validate the robustness against inaccurate pose extraction (a common issue in edge camera tracking), we introduce synthetic Gaussian noise $\mathcal{N}(0, \sigma^2)$ directly into the joint coordinates during testing. As illustrated in Fig. 4, when the noise level is increased to $\sigma = 0.1$, the standalone baseline suffers a catastrophic performance degradation of -10.96% (from 92.28% down to 81.32%). Conversely, MAG-KD demonstrates exceptional resilience, dropping merely -1.07% (from 94.44% to 93.37%). This compelling quantitative evidence verifies that our gating mechanism successfully filters out corrupted spatial topologies.

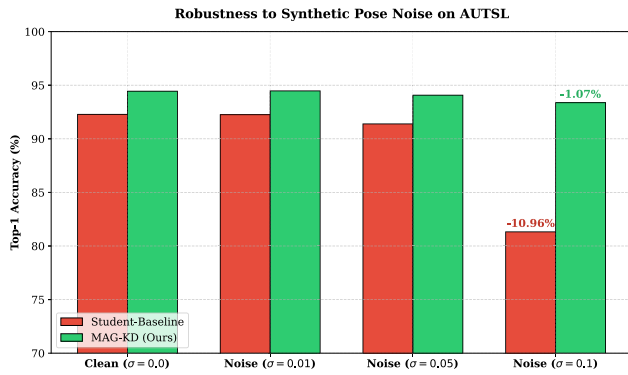


FIGURE 4. Robustness evaluation under different synthetic noise levels (σ) on the AUTSL dataset. MAG-KD maintains high accuracy even when severe spatial noise disrupts the baseline.

4) HYPERPARAMETER SENSITIVITY

We conduct an ablation study to assess the sensitivity of hyperparameters τ , α , and β , as visualized in Fig. 5. The results indicate that setting the distillation temperature $\tau = 4$ best balances the softening of the target distribution. Meanwhile, maintaining equal weighting ($\alpha = 1.0$, $\beta = 1.0$) for both logit and feature alignment yields the optimal convergence trajectory without causing gradient imbalances.

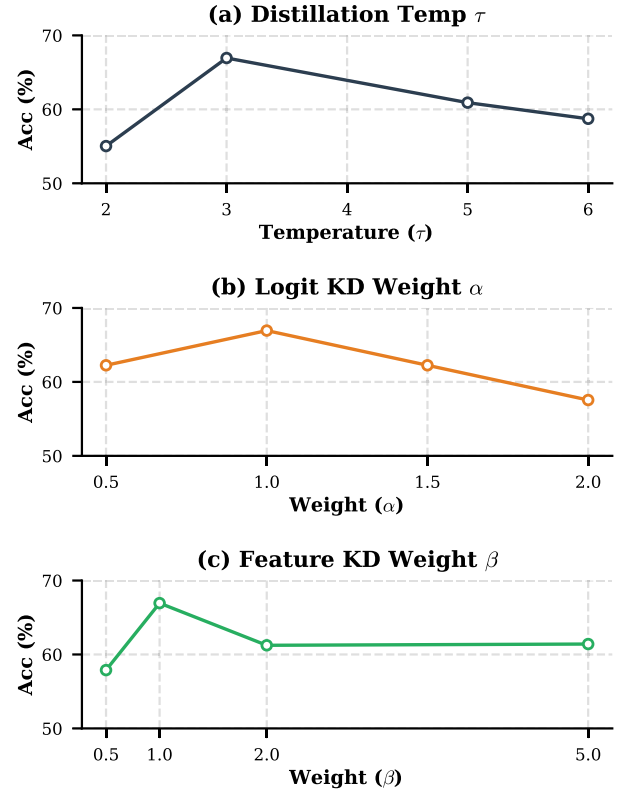


FIGURE 5. Ablation analysis on the distillation temperature (τ) and the loss balancing coefficients (α , β) on the WLASL-100 dataset.

F. QUANTITATIVE AND QUALITATIVE ERROR ANALYSIS

To explicitly quantify the practical impact of the proposed MAG-KD, we analyze the class-wise prediction dynamics of the student model compared to the baseline. While the overall system achieves a robust net gain of +81 correct samples (successfully rescuing 138 previously misclassified instances while introducing only 57 new errors), analyzing the performance variation across individual classes provides deeper insights. For visual clarity, as illustrated in Fig. 6, we highlight the top 15 sign language classes exhibiting the most extreme performance variations (specifically, the 10 most improved and the 5 most degraded classes).

Our framework introduces substantial accuracy improvements in the majority of ambiguous classes, with peak gains reaching an impressive +50.0% (*Government*) and +29.4% (*Alone*). Although the knowledge distillation process inevitably introduces minor prediction noise, causing slight performance drops in a smaller subset of classes (e.g., -21.4% in *Ataturk*), the magnitude of positive net gains significantly outweighs the degradations. Notably, the classes experiencing the highest improvements are predominantly characterized by complex, fine-grained hand movements, demonstrating that the correctness-aware mask successfully distills highly discriminative spatial-temporal features.

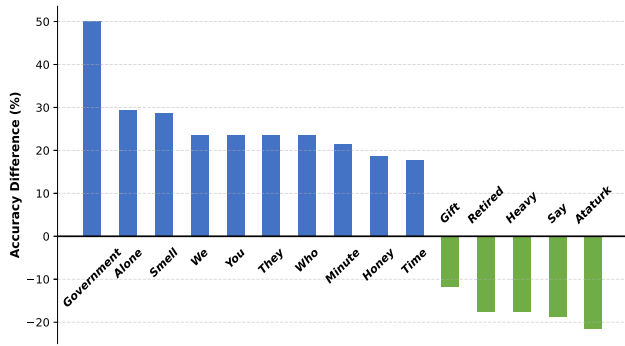


FIGURE 6. Quantitative error analysis on the AUTSL dataset. The chart highlights the top 15 classes with the most extreme accuracy variations (MAG-KD vs. Baseline). Blue bars represent the top 10 most improved sign language classes (explicitly labeled with their semantic meanings), while green bars show the 5 most degraded ones. The substantial magnitude of positive gains visually confirms the robustness of the distillation process.

To visually comprehend this enhancement, Fig. 7 provides a qualitative skeleton analysis. For instance, the ground truth action ‘Minute’ (Class 47) and the action ‘Do’ (Class 209) share highly similar macro-trajectories, as both involve placing one hand over the other arm. The primary distinguishing feature lies strictly in the fine-grained finger topology (e.g., a specific circular hand shape for ‘Minute’). When simulated tracking errors ($\sigma = 0.1$) disrupt these local finger joints into a corrupted spatial blob, the baseline model is completely deceived by the macro-movement and misclassifies the sequence as ‘Do’. In stark contrast, MAG-KD successfully filters out this topological noise and retains the true semantic meaning by relying on the robust, denoised features distilled from the teacher ensemble, accurately predicting the correct class.

G. SCALABILITY ANALYSIS ON ULTRA-LIGHTWEIGHT MODELS

To investigate the generalization capability and operating limits of the proposed framework, we extend our evaluation to the ultra-lightweight Student-S architecture containing approximately 72K parameters. This experiment aims to determine whether the proposed method remains effective when the student network suffers from severe capacity constraints. The comparative results presented in Table 5 reveal a distinct performance divergence compared to the standard experiments.

TABLE 5. Performance comparison on the 72K-Parameter Model (Student-S) on the AUTSL dataset.

Method	Top-1 Accuracy (%)
Baseline Student	83.57
Best Single Teacher	83.30
Multi-Teacher AVG	88.94
Proposed MAG-KD	77.69

While the proposed MAG-KD achieves state-of-the-art results on the 483K-parameter model, it experiences a

notable performance drop on this 72K model, yielding an accuracy of 77.69%, which falls below the standalone baseline of 83.57%. In contrast, the static Multi-Teacher Averaging strategy maintains robustness and achieves the highest accuracy of 88.94%. This phenomenon suggests that MAG-KD imposes an optimization overhead requiring a minimum representational capacity from the student. Consequently, when the model capacity is restricted to fewer than 100K parameters, the student struggles to learn the dynamic weighting policy alongside the main classification task, rendering simple static averaging a more effective strategy for such extreme edge scenarios.

H. REAL-WORLD DEPLOYMENT ON EDGE HARDWARE

To validate the practical feasibility of our proposed framework, we conducted a comprehensive evaluation on the NVIDIA Jetson Nano under the setup described in Section III-B. Table 6 summarizes the comparative performance between the Teacher ensembles and the Student models in terms of accuracy, inference speed, latency, model size, and energy efficiency. Preliminary experiments indicated that INT8 quantization on the Jetson Nano’s Maxwell architecture did not yield significant throughput gains but introduced non-negligible accuracy degradation. Therefore, FP16 precision and knowledge distillation act as complementary strategies in our pipeline to achieve optimal edge performance.

1) ANALYSIS OF TEACHER MODELS

As observed, while the Teacher models achieve high accuracy (up to 95.35% for SL-GCN), they are computationally prohibitive for edge deployment. Specifically, SL-GCN and TD-GCN suffer from extremely high latency (~152–153 ms), resulting in a throughput of only ~6.5 FPS, which falls far below the real-time requirement (typically 30 FPS). Even the most efficient Teacher, CTR-GCN, only reaches 11.53 FPS. Furthermore, their energy efficiency is notably low, averaging between 0.97 and 1.75 FPS/W, which indicates a heavy computational burden on the embedded GPU.

2) PERFORMANCE OF MAG-KD

In contrast, our proposed Student-M (MAG-KD) demonstrates a superior trade-off between accuracy and efficiency. This optimal balance is visually corroborated by the Accuracy-Latency chart in Fig. 8. By leveraging the TensorRT FP16 optimization and the proposed distillation method, MAG-KD reduces the inference latency to 30.19 ms, effectively achieving a throughput of 33.13 FPS. This result confirms that our method satisfies the strict real-time constraint of standard video streams (>30 FPS).

Compared to the best Teacher (SL-GCN), MAG-KD provides a $5.1\times$ speedup and reduces the model storage size by $3\times$ (from 32.44 MB to 10.69 MB), with a minor accuracy trade-off of 1.10%. Moreover, regarding energy consumption, MAG-KD operates at a lower power draw

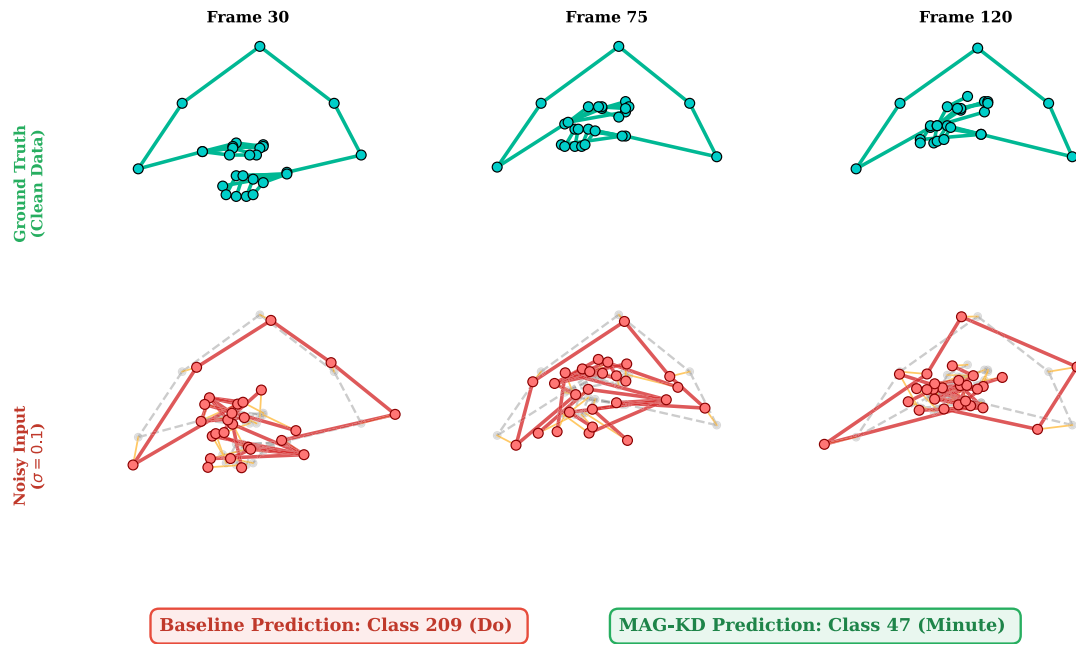


FIGURE 7. Qualitative skeleton visualization on the AUTSL dataset. The bottom row displays simulated tracking errors (noisy input with displacement ghosting). While the baseline is deceived by the corrupted finger joints and misclassifies ‘Minute’ as ‘Do’, MAG-KD correctly classifies the action by leveraging robust distilled features.

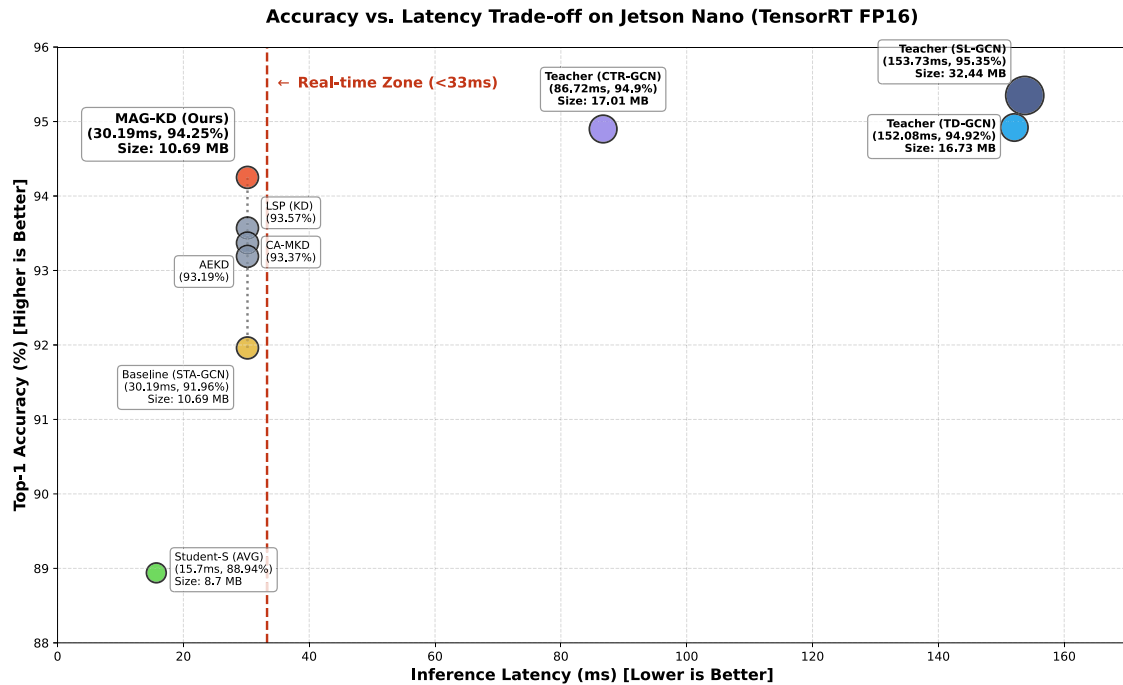


FIGURE 8. Accuracy vs. Latency trade-off on NVIDIA Jetson Nano. At an identical latency footprint (~ 30.19 ms), our proposed MAG-KD (red circle) achieves the optimal balance, delivering SOTA accuracy (94.25%) and outperforming all other competing KD baselines while strictly satisfying real-time constraints (< 33 ms).

(6.23 W) and achieves an energy efficiency of 5.32 FPS/W, which is approximately $5.5\times$ higher than that of the SL-GCN baseline.

3) EXTREME EFFICIENCY CASE

Additionally, the smaller variant, Student-S, pushes the speed capability further to 63.70 FPS with an ultra-low

TABLE 6. Deployment efficiency on edge hardware (Jetson Nano) evaluated on the AUTSL dataset. All models are compiled using TensorRT with FP16 precision. The Δ column denotes the absolute accuracy improvement relative to the baseline. The proposed MAG-KD model is highlighted in bold.

Model	Params (M)	FLOPs (G)	Acc (%)	Δ (%)	Speed (FPS)	Latency (ms)	Size (MB)	RAM (MB)	GPU (%)	Power (W)	Eff. (FPS/W)
<i>Teacher Models (Baselines)</i>											
SL-GCN	4.87	3.53	95.35	-	6.50	153.73	32.44	1591.0	99.7	6.69	0.97
TD-GCN	1.42	2.09	94.92	-	6.58	152.08	16.73	1195.3	99.6	6.50	1.01
CTR-GCN	1.49	2.20	94.90	-	11.53	86.72	17.01	1198.9	99.1	6.60	1.75
<i>Student Models (Ours)</i>											
Baseline (STA-GCN)	0.48	0.55	91.96	0.00	33.13	30.19	10.69	1187.0	97.5	6.20	5.34
Student-M (MAG-KD)	0.48	0.55	94.25	+2.29	33.13	30.19	10.69	1187.8	97.8	6.23	5.32
Student-S (AVG)	0.07	0.09	88.94	-3.02	63.70	15.70	8.70	1184.0	94.5	6.65	9.58

latency of 15.70 ms. While its accuracy drops to 88.94%, it offers an impressive efficiency of 9.58 FPS/W, making it a viable candidate for extremely resource-constrained scenarios where speed is the primary priority.

In conclusion, the experimental results on the Jetson Nano validate that the proposed MAG-KD framework successfully compresses heavy GCN ensembles into a lightweight, energy-efficient model capable of real-time sign language recognition on edge devices.

4) SYSTEM-LEVEL PROJECTIONS AND END-TO-END LIMITATIONS

While our evaluation fundamentally focuses on the isolated skeleton-based recognition phase, a real-world Sign Language Recognition (SLR) system operates as an end-to-end sequential pipeline. This invariably involves a heavy upstream stage: Human pose extraction from raw RGB videos.

Extensive deployments reported by Alves et al. [48] and Pääkkönen and Pakkala [49] demonstrate that executing robust pose estimation (e.g., MediaPipe) on edge devices like the Jetson Nano incurs a massive latency penalty, typically occupying ~ 77 ms per frame. This upstream bottleneck is precisely the scientific justification for demanding an ultra-lightweight downstream recognition module. If a high-capacity model like SL-GCN (~ 153 ms latency) were deployed end-to-end, the total system latency would exceed 230 ms per frame (< 4.5 FPS), paralyzing real-time interactions. By strictly compressing the recognition latency to ~ 30.19 ms via MAG-KD, the projected end-to-end latency of the entire system is maintained at ~ 107.19 ms (≈ 9.3 FPS). This calculation transparently confirms that our model's compactness is a strict prerequisite for viable end-to-end edge SLR systems.

IV. DISCUSSION

In this section, we interpret the experimental findings, focusing on the unique challenges of distilling graph-based models and the theoretical implications of the capacity-adaptability trade-off.

A. MITIGATING NEGATIVE TRANSFER IN SKELETON-BASED SLR

A critical distinction between standard image-based classification and skeleton-based SLR lies in the information

density of the input data. Image data typically exhibits high spatial redundancy, where minor noise or artifacts often have a negligible impact on semantic interpretation. In contrast, skeleton data is highly sparse, consisting of only 27 joint coordinates per frame. Consequently, a single noisy joint caused by self-occlusion or estimation errors can drastically alter the semantic meaning of a sign. Our ablation study presented in Table 4 demonstrates that removing the *Correctness Mask* causes the student's accuracy to drop significantly to 85.30%, which is even lower than the baseline. This confirms that *Negative Transfer*, where the student mimics incorrect predictions from the teacher, is a severe issue in GCN distillation. The Correctness Mask functions as a "Hard Oracle," ensuring that the student only distills knowledge from teachers that have correctly reasoned about the input graph. This finding underscores that for sparse graph-based tasks, the *quality* of supervision signals is far more critical than the *quantity*.

B. THE CAPACITY-ADAPTABILITY TRADE-OFF

The performance divergence observed between the Student-M with 483K parameters and Student-S with 72K parameters in Section III-G aligns with the capacity gap phenomenon discussed in knowledge distillation literature [50], [51]. Our proposed Adaptive Gating Network relies on the student's feature maps to dynamically calculate the importance weight for each teacher. For the Student-M, the representational capacity is sufficient to extract discriminative features, enabling the gating mechanism to effectively identify and prioritize the most relevant teacher. However, for the ultra-lightweight Student-S, the severe parameter reduction leads to a representational bottleneck. As noted by Cho et al. [50], when the student lacks sufficient capacity, it fails to capture the complex decision boundaries required to guide the adaptation process. Consequently, the gating network receives sub-optimal features, leading to unstable weight assignments. In such extremely low-capacity regimes, static ensembles such as AVER, which do not depend on the student's features, offer better stability, consistent with observations by Mirzadeh et al. [51].

Based on these insights, we recommend MAG-KD for standard edge devices such as the Jetson Nano and Xavier, where the model capacity is moderate, approximating 0.5M parameters. For extreme edge scenarios, including microcontrollers requiring ultra-compressed models with fewer than

0.1M parameters, static averaging remains a safer and more robust choice.

C. LIMITATIONS AND FUTURE WORK

While MAG-KD achieves state-of-the-art performance, we acknowledge several primary limitations. First, the efficacy of the proposed Adaptive Gating is contingent upon the discriminative power of the student's features. As analyzed in Section IV-B, in ultra-low capacity regimes with fewer than 0.1M parameters, the student suffers from a representational bottleneck, failing to guide the gating mechanism effectively. Consequently, performance in this specific tier degrades compared to static averaging strategies. Second, our Correctness Mask relies entirely on ground-truth labels during the training phase, acting as an "Oracle." This strict dependency currently prevents the direct extension of our framework to semi-supervised or self-distillation paradigms, where ground-truth labels are sparse or unavailable. Third, our current evaluation focuses exclusively on isolated gesture recognition. Real-world sign language involves continuous streams with complex co-articulation effects, which our decoupled masking mechanism is not yet equipped to handle without robust sequence-alignment architectures (e.g., CTC). Finally, our framework operates exclusively on the skeleton modality. Although this design ensures high computational efficiency, it inherits the dependency on upstream pose estimation quality. In challenging real-world scenarios characterized by severe occlusions or low-light conditions, pose estimation failures will inevitably compromise recognition accuracy.

In future work, we aim to bridge the capacity gap by investigating lightweight, parameter-free gating mechanisms or curriculum distillation strategies tailored for ultra-compressed models. Additionally, we plan to extend the framework to accommodate continuous sign language translation (CSLT) streams and explore self-supervised consensus mechanisms among teachers to dynamically estimate pseudo-ground-truths, thereby removing the reliance on the oracle mask. Furthermore, to mitigate the reliance on pose quality and alleviate the upstream latency bottleneck, we plan to extend the framework to a cross-modal distillation setting and explore end-to-end joint optimization with ultra-lightweight pose estimators. By leveraging robust Teacher signals from RGB or Depth modalities during training, we aim to enhance the Student's resilience against noisy skeleton inputs without increasing inference costs.

V. CONCLUSION

In this work, we addressed the critical challenge of deploying robust skeleton-based Sign Language Recognition systems on resource-constrained edge devices, where inherent pose estimation noise often hinders standard distillation methods. To overcome the limitation of negative transfer, we introduced the Mask-Guided Adaptive Knowledge Distillation (MAG-KD) framework. By strategically decoupling feature

alignment from logit distillation and enforcing a Correctness-aware Mask, our approach ensures that the student model learns exclusively from reliable teacher supervision while dynamically synthesizing knowledge from a heterogeneous ensemble (SL-GCN, CTR-GCN, TD-GCN).

Extensive experiments on the AUTSL and WLASL-100 datasets consistently validate the superiority and cross-language generalizability of MAG-KD. The empirical results demonstrate that our lightweight student model significantly outperforms both the static baseline and state-of-the-art knowledge distillation methods, while requiring only a fraction of the parameters compared to the high-capacity teachers. Furthermore, comprehensive hardware profiling on the NVIDIA Jetson Nano confirms that the proposed framework successfully bridges the theoretical-practical gap, achieving real-time inference alongside highly efficient power consumption.

Despite these advancements, the proposed framework exhibits certain limitations. First, MAG-KD relies strictly on ground-truth labels to formulate the correctness mask, which restricts its direct application in semi-supervised learning paradigms. Second, as an exclusively skeleton-based approach, the system's end-to-end performance remains inherently bottlenecked by the latency and accuracy of upstream pose extractors in challenging environments. Therefore, future work will focus on developing self-supervised consensus mechanisms to estimate pseudo-labels, investigating parameter-free gating for extreme edge scenarios, and exploring cross-modal distillation to alleviate the strict dependency on upstream skeleton extraction.

DATA AVAILABILITY STATEMENT

The AUTSL and WLASL datasets used in this study are publicly available in Ref. [44] and Ref. [45], respectively. The source code and trained models generated during the current study are available from the corresponding author upon reasonable request.

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